Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science, Probability for Data Science, Introduction to Urban Data Analytics; *In progress:* Introduction to Machine Learning, Data Engineering

Technical Skills

Languages: Python, SQL, Java, TypeScript

Product: User Research, User Interviews, Discovery Read-outs, Documentation, Cross-Functional Collaboration

Data & ML: pandas, NumPy, scikit-learn, XGBoost, PyTorch, PostgreSQL, ArcGIS Online, Jupyter

Engineering: FastAPI, React, Supabase, Claude API, Apify, Factory Droid SDK, LaTeX, REST APIs, AWS S3, pytest

Experience

Fung Fellowship | Fung Fellow, Health + Innovation | Berkeley, CA **Aug 2026 - Present**

- Mentored by The After Collective on a team building a free guide for wildfire survivors; DRI of the week for **6** weeks.
- Led **4** interviews and ran **3** of **5** prototype sessions, focusing us on whoever runs the insurance claim.
- Built the prototype; rebuilt its start screen around the letter after users missed the reply helper.

Guang Labs | Engineering Intern | Remote **May 2026 - Aug 2026**

- Built a service for an AI video startup that sorts videos and images into **6** content types, so the platform can handle each type differently.
- Added a confidence check that retries uncertain results on long videos, and stored each label with its confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that tells users when their video finishes processing or fails, instead of leaving them waiting on an idle screen.

IDX Exchange | Data Science Intern | Remote **Jun 2026 - Aug 2026**

- Contributed to a California home-price machine learning model trained on **71K** single-family sales; the final model's typical estimate was off by **8.6%** (R^2 of **0.88**).
- Traced an unusable first result to a data leak, fixed it, and brought accuracy back to a usable range.
- Measured accuracy by price tier and across **161** school districts, showing estimates were least reliable for luxury homes (**13%** error vs. **6.5%** mid-priced) and resort markets.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Supabase, Apify, Factory Droid* GitHub

- Co-designed a transfer planner: a **FastAPI** backend that matches courses to official ASSIST agreements with deterministic rules (minimum grades, agreement years), a **React** and **TypeScript** interface, and **Supabase**.
- **Claude** reads transcripts and class schedules that students confirm and answers with numbered citations that code validates; **Apify** scrapes campus pages and a **Factory Droid** agent repairs only broken scraper scripts, in a sandbox.
- Tested with **10** transfer students, who said it would have helped during their own transfer.

Resume Tailor | *Python, FastAPI, LaTeX, Claude* GitHub

- Built with **Claude**: a **Python** and **FastAPI** app that tailors a **LaTeX** resume to a job posting.
- **Claude API** rewrites must quote my own records; code rejects invented numbers and checks page fit and ATS parsing.

Next Step (Fung Fellowship) | *JavaScript, Cloudflare, Claude* GitHub

- Built with **Claude**: a site that helps survivors write one clear insurance letter, grounded in **15** cited sources.

Leadership & Involvement

UC Berkeley CDSS | Dean's Cabinet Transfer Representative | Berkeley, CA **Sep 2026 - Present**

- Advise CDSS leadership on the student experience for CDSS majors.

UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA **Aug 2026 - Present**

- Give resume feedback in a careers seminar, and run exam reviews and practice worksheets for CS and data science classes for transfer students.

UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.